

Designing a Scalable Enterprise VPC with Multi-Sized Subnets:-

Created a VPC :-

Go to **VPC → Your VPCs → Create VPC**

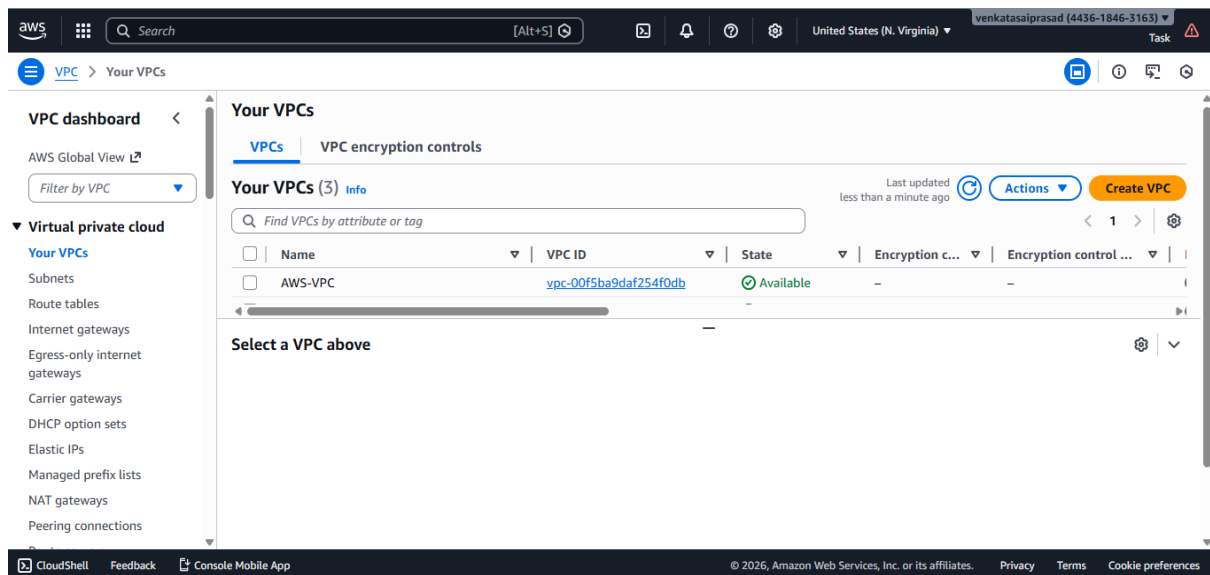
Name tag: Enterprise-VPC

IPv4 CIDR: 10.0.0.0/16

IPv6: None (optional)

Tenancy: Default

Click **Create VPC** 



Subnets Design:-

Subnet Name	Purpose	Required IPs	CIDR
Admin	Bastion / Ops	~256	10.0.62.0/24
Edge	Ingress / LB	~512	10.0.60.0/23
Web	Web Tier	~1024	10.0.56.0/22
App	App Tier	~2048	10.0.48.0/21
Platform	Containers / Tools	~4,096	10.0.32.0/20
Shared	Large Internal Services	~8192	10.0.0.0/19

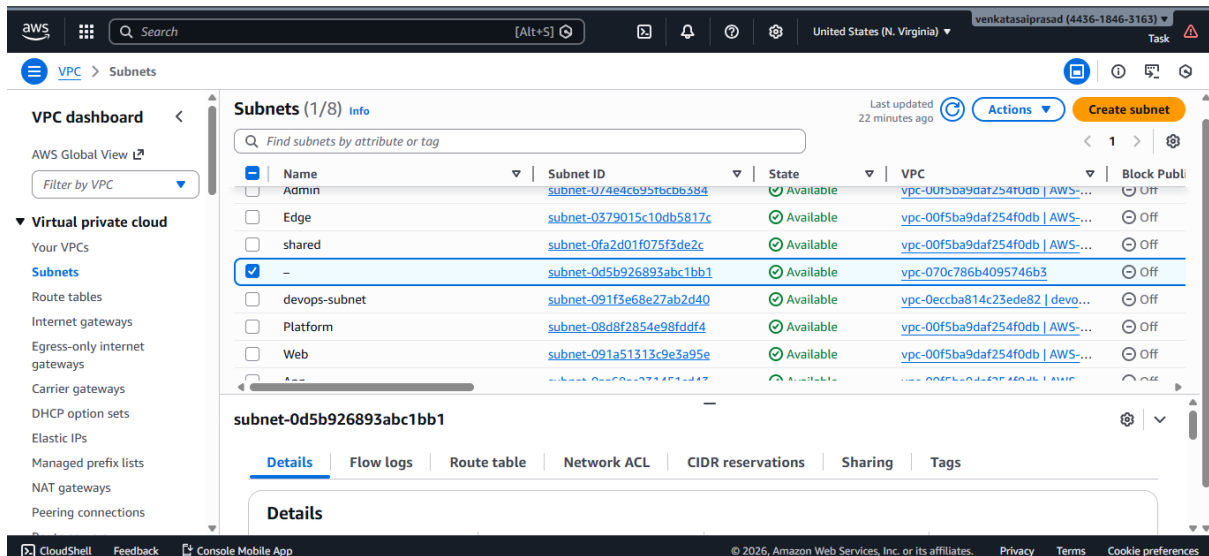
Go to **VPC → Subnets → Create subnet**

Select **VPC = Enterprise-VPC**

Enter subnet details (Name, CIDR, AZ)

Click **Create subnet**

Repeat for all 6 subnets ✓

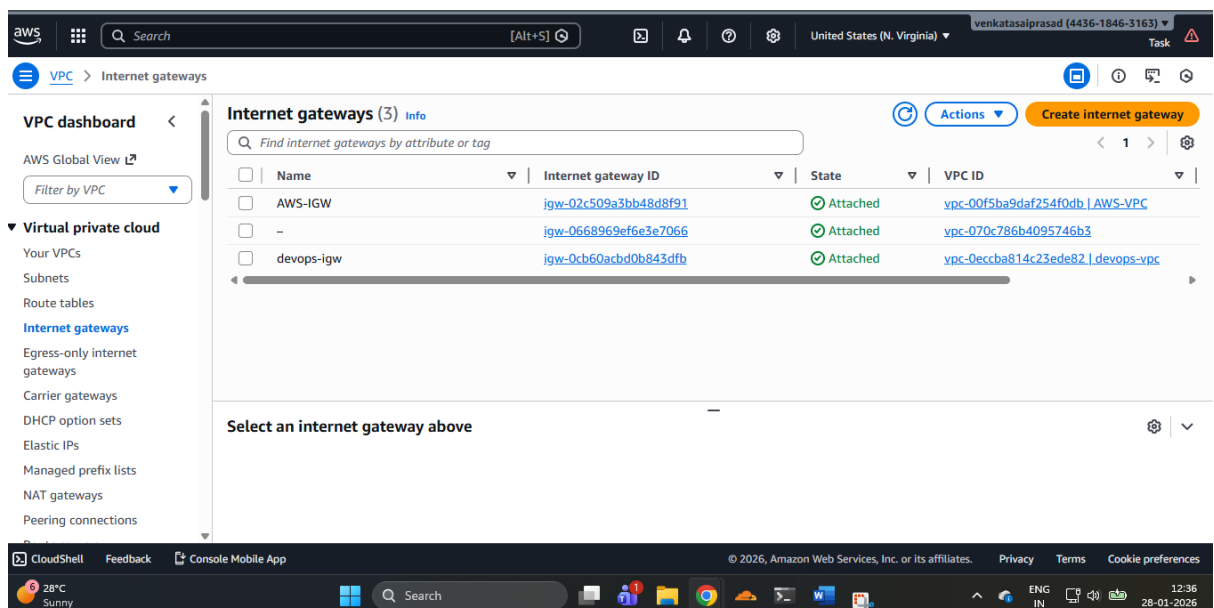


1. Go to **VPC** → **Internet Gateways** → **Create internet gateway**

2. Name: Enterprise-IGW

3. Click **Create**

4. Select IGW → **Actions** → **Attach to VPC** → **Enterprise-VPC**



Step 4: Create Route Tables

4a. Public Route Table (Admin + Edge)

1. Go to **VPC → Route Tables → Create route table**
2. Name: Public-RT
3. VPC: Enterprise-VPC
4. Click **Create**
5. Select Public-RT → Routes → Edit routes → Add route
 - Destination: 0.0.0.0/0
 - Target: IGW → Enterprise-IGW
6. Keep default local route 10.0.0.0/16 ☒

4b. Private Route Table (Web, App, Platform, Shared)

1. Go to **Route Tables → Create route table**
2. Name: Private-RT
3. VPC: Enterprise-VPC
4. Routes: Only local 10.0.0.0/16 ☒

The screenshot shows the AWS Management Console interface for the 'Route tables' section. The left sidebar contains the 'VPC dashboard' and a list of VPC resources including 'Virtual private cloud', 'Your VPCs', 'Subnets', 'Route tables', 'Internet gateways', 'Egress-only internet gateways', 'Carrier gateways', 'DHCP option sets', 'Elastic IPs', 'Managed prefix lists', 'NAT gateways', and 'Peering connections'. The main content area displays a table of route tables for the selected VPC. The table has columns for 'Name', 'Route table ID', 'Explicit subnet associations', 'Edge associations', 'Main', and 'VPC'. The 'private-rt' is highlighted, showing it is associated with 4 subnets. The 'public-rt' is also visible, associated with 2 subnets. The interface includes a search bar, a filter by VPC dropdown, and a 'Create route table' button. The bottom of the screen shows the Windows taskbar with the date 28-01-2026.

Name	Route table ID	Explicit subnet associations	Edge associations	Main	VPC
-	rtb-082c0b419bd238e85	-	-	Yes	vpc-0
private-rt	rtb-0e01ac5f2abf2e13e	4 subnets	-	No	vpc-0
devops-public-rt	rtb-0d0ef11a038c56c65	subnet-091f3e68e27ab2...	-	No	vpc-0
-	rtb-0f29b9769e20fe856	-	-	Yes	vpc-0
public-rt	rtb-078868d9a01849bac	2 subnets	-	No	vpc-0
-	rtb-04fc265daa95d13ad	-	-	Yes	vpc-0

The top screenshot shows the AWS Management Console interface for the 'Route tables' section. The left sidebar displays the 'VPC dashboard' with a list of VPC resources. The main content area shows the 'Route tables (1/6)' list. The 'public-rt' route table is selected, and its 'Explicit subnet associations (2)' are displayed in a table below.

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
Admin	subnet-074e4c695f6cb6384	10.0.62.0/24	-
Edge	subnet-0379015c10db5817c	10.0.60.0/23	-

The bottom screenshot shows the same AWS Management Console interface, but the 'private-rt' route table is selected. Its 'Explicit subnet associations (4)' are displayed in a table below.

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
shared	subnet-0fa2d01f075f3de2c	10.0.0.0/19	-
Platform	subnet-08d8f2854e98fddf4	10.0.32.0/20	-
Web	subnet-091a51313c9e3a95e	10.0.56.0/22	-
App	subnet-0aa68ac231451cd43	10.0.48.0/21	-

Launch Test EC2 Instances

1. Launch **1 EC2 per subnet** (use Amazon Linux 2 AMI)
2. Public subnets: assign **public IP**
3. Private subnets: **no public IP**

Security Groups: allow ICMP (ping) and SSH for testing

aws Search [Alt+S] United States (N. Virginia) venkatasaiprasad (4436-1846-3163) Task

EC2 > Instances > Launch an instance

Network settings

VPC - required

vpc-00f5ba9daf254f0db (AWS-VPC)

Subnet

subnet-074e4c695f6cb6384 Admin

Auto-assign public IP

Enable

Firewall (security groups)

Create security group Select existing security group

Security group name - required

launch-wizard-3

Summary

Number of instances

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read more

ami-0b6c6ebcd2801a5cb

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel Launch instance Preview code

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters.

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EC2 > Instances

EC2

Dashboard

EC2 Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

Instances (1/2)

Find Instance by attribute or tag (case-sensitive)

All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability
☐	kops-commands	i-0560194c93aa66810	Stopped	c7i-flex.large	-	View alarms +	us-east-1a
☒	Test-instance	i-03af28dd64a5c69c3	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1a

i-03af28dd64a5c69c3 (Test-instance)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary

Instance ID	Public IPv4 address	Private IPv4 addresses
i-03af28dd64a5c69c3		

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Go to the terminal

→ sudo apt update

```
ubuntu@ip-10-0-62-92:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1700 kB]
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [314 kB]
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.0 kB]
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1519 kB]
Get:17 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [310 kB]
Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:19 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [31.6 kB]
Get:20 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2506 kB]
Get:21 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [573 kB]
Get:22 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
```

→sudo apt install nginx -y

```
ubuntu@ip-10-0-62-92:~$ sudo apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 62 not upgraded.
Need to get 564 kB of archives.
After this operation, 1596 kB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.5 [43.4 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.5 [520 kB]
Fetched 564 kB in 0s (19.3 MB/s)
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.5_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.5) ...
Preparing to unpack .../nginx_1.24.0-2ubuntu7.5_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.5) ...
Setting up nginx-common (1.24.0-2ubuntu7.5) ...
Setting up nginx (1.24.0-2ubuntu7.5) ...
```

→sudo systemctl start nginx

→sudo systemctl enable nginx

→sudo systemctl status nginx

```
ubuntu@ip-10-0-62-92:~$ sudo systemctl start nginx
ubuntu@ip-10-0-62-92:~$ sudo systemctl enable nginx
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nginx
ubuntu@ip-10-0-62-92:~$ systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-01-28 07:15:28 UTC; 3min 34s ago
     Docs: man:nginx(8)
    Main PID: 1746 (nginx)
      Tasks: 3 (limit: 1008)
     Memory: 2.4M (peak: 5.3M)
        CPU: 27ms
    CGroup: /system.slice/nginx.service
            └─1746 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
               └─1748 "nginx: worker process"
                 └─1749 "nginx: worker process"

Jan 28 07:15:28 ip-10-0-62-92 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server:
Jan 28 07:15:28 ip-10-0-62-92 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server:
lines 1-15/15 (END)
```

Test NGINX

Open a web browser and go to your server's IP address:

<http://18.209.248.1/>

